

Jose Ortiz Flores

CONTACT

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HONORS & SCHOLARSHIPS

- Fulbright García-Robles Scholarship**
Graduate Studies in the USA | 2021
- ISAP Scholarship for Exchange Studies**
RWTH Aachen University | 2020- 2021
- ERASMUS Scholarship**
Jönköping University | 2020
- ITESM Scholarship**
Undergraduate Studies in Mexico | 2018- 2021

CERTIFICATIONS

- Guidance and Control of Hypersonic Vehicles**
AIAA | 2025
- Fundamentals of Atmospheric Entry**
White Eagle Aerospace | 2025

LANGUAGES

- Spanish** | Native
- English** | Fluent

SKILLS

SOFTWARE

- | | |
|----------|-----------|
| MATLAB | GMAT |
| Simulink | STK |
| Python | Freeflyer |
| Linux OS | Jira |
| GitHub | Excel |

PERSONAL

- Analytical Research
- Technical communication
- Cross-functional collaboration
- Systems-level problem-solving
- Mentoring & Team Development

EDUCATION

- M.S. in Aeronautics and Astronautics | 2022 - 2024**
Purdue University | Indiana, U.S.A.
 - Major in Astrodynamics & Minor in Autonomy and Control (GPA 3.8/4)
 - Graduate Teaching Assistant for the College of Engineering Honors Program
- B.S. in Mechanical and Electrical Engineering | 2016 - 2021**
Instituto Tecnológico y Estudios Superiores de Monterrey (ITESM) | Mexico City, Mexico
 - Summa Cum Laude (2nd in class with a 96/100 cumulative average)
- Undergraduate Study Abroad | 2021**
Rheinisch-Westfälische Technische Hochschule (RWTH) | Aachen, Germany
 - Completed graduate-level coursework in: Aerothermal Design of Space Vehicles, Spacecraft Systems, Space Propulsion, Space Flight Dynamics, and Structural Design of Spacecraft
- Undergraduate Study Abroad | 2020**
Jönköping University | Jönköping, Sweden
 - Completed graduate-level courses in Mechanical and Materials Engineering

EXPERIENCE

- Embry-Riddle Aeronautical University | Visiting Assistant Professor**
Florida, U.S.A. | Aug 2024 - May 2026
 - Professor of Space Mechanics, Spacecraft Attitude Dynamics, Space Environmental Applications, and Spacecraft Systems
 - Developed and implemented complementary and Kalman filters, an LQR controller, and path-planning algorithms for a Parrot Mambo minidrone using MATLAB Simulink
 - Utilized a real 3U CubeSat platform to demonstrate satellite detumbling using a reaction wheel
- CISTAR – NSF Engineering Research Center | Summer Research Program Mentor**
Indiana, U.S.A. | 2023- 2024
 - Mentored participants in CISTAR’s summer research programs, including REM, REU, RET, and YS
- Ford Motor Company | Hardware-in-the-Loop Junior Engineer**
Mexico City, Mexico | 2022
 - Tested automotive cluster and dashboard features in accordance with established test protocols, ensuring both hardware and software performance met functional requirements
- O&F Integrated Specialized Engineering | Lead Systems Engineer**
Mexico City, Mexico | 2015- 2021
 - Led the installation, integration, commissioning, and troubleshooting of access control, fire suppression, CCTV, and emergency power systems through project delivery.

LEADERSHIP EXPERIENCE

- ICARUS Sounding Rocket Program | Team Advisor**
Florida, U.S.A. | 2024- 2026
 - Advised a team of 100+ undergraduate students at Embry-Riddle on designing, building, and launching a sounding rocket aimed at surpassing the Kármán line
- Student Team of Aeronautics (SAIA) | Team Captain**
Mexico City, Mexico | 2017- 2020
 - Led an interdisciplinary team of 40 students in the design, construction, and flight testing of an RC cargo aircraft for national and international SAE competitions.
 - Achieved two consecutive third-place finishes.

RESEARCH EXPERIENCE

- Purdue University**
Indiana, U.S.A. | 2023- 2024
 - Conducted research on atmospheric reentry guidance performance, employing sequential convex programming and multiple trust-region strategies. Presented findings at the International Planetary Probe Workshop (IPPW) 2024
- Instituto Tecnológico y Estudios Superiores de Monterrey**
Mexico City, Mexico | 2021
Advisors: Dr. Luis M. Ibarra-Moyers (ITESM) and Dr. Jorge A. Ferrer-Pérez (UNAM)
 - Developed a thermal model for a thermal vacuum chamber to support future nanosatellite research by enabling compliance with heating and cooling requirements under ISO 19683 standards
- King Abdullah University of Science and Technology (KAUST)**
Thuwal, Saudi Arabia | 2021
Advisors: Dr. Francisco Hernández, Dr. Minh Bau Long, and Dr. Hong G. Im
 - Developed an in-house direct numerical simulation (DNS) code for high-pressure, high-Reynolds-number reacting flow problems, demonstrating performance through simulations of canonical combustor configurations.